Morse Lemma. (Normal forms)

Diffeomorphisms 微分同胚　　光滑：无限可微.

bijective 双射 mapping : $F: U \to V$, F^{-1} denoting the inverse mapping.

定义:

F is C^k-diffeomorphism $(k \geq 1)$ if $F \in C^k(U, V)$ and $F^{-1} \in C^k(V, U)$

$\downarrow$ Continuous differentiable.　C^1: 一阶连续可微 (导数存在且连续).　C^0: 连续.

Optimality Conditions 最优性条件

优化问题中：如果 x^* 是局部最优解

它必须满足什么条件?

本质："一个解是最优的，需要满足某些方程/条件"　{ 极值问题的必要条件 拉格朗日乘子法.

思想：calculus of variations 变分法

the method of Lagrange multipliers : 寻找 local maxima and minima.

(Multivariable Calculus)

例子：maximize $f(x, y) = xy$

Subject to $3x + y = 720$　　$g(x, y) = 3x + y - 720 = 0$

$$F(x, y, \lambda) = f(x, y) + \lambda g(x, y) = xy + \lambda(3x + y - 720)$$

$$F_x(x, y, \lambda) = 0 \Rightarrow y + 3\lambda = 0$$

$$F_y(x, y, \lambda) = 0 \Rightarrow x + \lambda = 0 \qquad \Rightarrow \lambda = -120 \quad x = 120, \; y = 360.$$

$$F_\lambda(x, y, \lambda) = 0 \Rightarrow 3x + y - 720 = 0$$

The solutions are critical points of F.

PCA 优化最大方差 : $\max u^T S u$

Subject to $\|u\| = 1$

$$F(u, \lambda) = u^T S u + \lambda(u^T u - 1)$$

$$F_u = u^T S + \lambda u^T = 0 \qquad u^T S = \boxed{-\lambda} u^T \Rightarrow S^T u = -\lambda u$$

对称 $\Rightarrow S u = -\lambda u$

$\rightarrow$ eigenvalue.

$$F_\lambda = u^T u - 1 = 0$$